ADBMS Project

Group – 29

Schema Documentation

Project Title: Vehicle Subscription(Rental) Management Platform

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DATA DICTIONARY

1) **Vehicle**

Column	Data Type	Constraints	Description
id	INT	PK,	Unique vehicle identifier
type	VARCHAR(50)	NOT NULL	Vehicle type (Sedan, SUV, etc.)
registration_year	YEAR	NOT NULL, CHECK (registration_year >= 1980)	Manufacturing year
make	VARCHAR(50)	NOT NULL	Brand (Toyota, Honda, etc.)
model	VARCHAR(50)	NOT NULL	Model name
transmission	VARCHAR(20)	NOT NULL, CHECK (transmission IN ('Manual', 'Automatic'))	Transmission type
mileage	INT	NOT NULL, CHECK (mileage >= 0)	Total mileage

Indexes

- PRIMARY KEY (id) for fast lookups
- INDEX idx_vehicle_make_model (make, model) for Searching/filtering vehicles
- INDEX idx_vehicle_type (type) for Filter by vehicle category

2) **Customer**

Column	Data Type	Constraints	Description
customer_id	INT	PK, AUTO_INCREMENT	
driver_id	INT	FK → Driver(driver_id), UNIQUE	
name	VARCHAR(100)	NOT NULL	
email	VARCHAR(100)	NOT NULL, UNIQUE	
phone_number	VARCHAR(20)	NOT NULL, UNIQUE	
address	VARCHAR(255)	NOT NULL	
license_number	VARCHAR(50)	NOT NULL, UNIQUE	

Indexes

- UNIQUE(email)

- UNIQUE(phone_number)
- UNIQUE(license_number)
- INDEX idx_customer_driver (driver_id) for join optimization

3) Driver

Column	Data Type	Constraints
driver_id	INT	PK, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	UNIQUE
phone_number	VARCHAR(20)	UNIQUE
address	VARCHAR(255)	NOT NULL
license_number	VARCHAR(50)	NOT NULL, UNIQUE
available	BOOLEAN	NOT NULL DEFAULT TRUE

Indexes

- INDEX idx_driver_available (available) for finding available drivers quickly
- UNIQUE(license_number)

4) Rental_Booking

Column	Data Type	Constraints
booking_id	INT	PK, AUTO_INCREMENT
customer_id	INT	FK → Customer(customer_id), NOT NULL
vehicle_id	INT	FK → Vehicle(id), NOT NULL
driver_id	INT	FK → Driver(driver_id)
booking_status	VARCHAR(20)	NOT NULL, CHECK (booking_status IN ('Pending','Confirmed','Completed','Cancelled'))
pickup_date	DATETIME	NOT NULL
return_date	DATETIME	NOT NULL, CHECK (return_date > pickup_date)

Indexes

- INDEX idx_booking_customer (customer_id)
- INDEX idx_booking_vehicle (vehicle_id)
- INDEX idx_booking_dates (pickup_date, return_date) for Availability checks
- INDEX idx_booking_status (booking_status)

Rationale:
These indexes support common queries like:

- Booking history
- Checking vehicle availability
- Filtering by status

5) Maintenance

Column	Data Type	Constraints
maintenance_id	INT	PK, AUTO_INCREMENT
vehicle_id	INT	FK, Vehicle(id), NOT NULL
service_date	DATE	NOT NULL
next_service_mileage	INT	CHECK (next_service_mileage >= 0)
service_type	VARCHAR(100)	NOT NULL

Indexes

- INDEX idx_maintenance_vehicle (vehicle_id)
- INDEX idx_maintenance_date (service_date)

6) Insurance

Column	Data Type	Constraints
policy_id	INT	PK, AUTO_INCREMENT
vehicle_id	INT	FK → Vehicle(id), NOT NULL
provider	VARCHAR(100)	NOT NULL
start_date	DATE	NOT NULL

Column	Data Type	Constraints
expiry_date	DATE	NOT NULL, CHECK (expiry_date > start_date)
premium_amount	DECIMAL(10,2)	NOT NULL, CHECK (premium_amount > 0)

Indexes

- INDEX idx_insurance_vehicle (vehicle_id)
- INDEX idx_insurance_expiry (expiry_date) for track expiring policies

7) Damage_Report

Column	Data Type	Constraints
damage_id	INT	PK, AUTO_INCREMENT
booking_id	INT	FK, Rental_Booking(booking_id), NOT NULL
policy_id	INT	FK, Insurance(policy_id)
damage_description	TEXT	NOT NULL
damage_part	VARCHAR(100)	NOT NULL
repair_cost	DECIMAL(10,2)	CHECK (repair_cost >= 0)
inspection_cost	DECIMAL(10,2)	CHECK (inspection_cost >= 0)
isResolved	BOOLEAN	DEFAULT FALSE

Indexes

- INDEX idx_damage_booking (booking_id)
- INDEX idx_damage_policy (policy_id)
- INDEX idx_damage_resolved (isResolved)

8) Purchase

Column	Data Type	Constraints
purchase_id	INT	PK, AUTO_INCREMENT
vehicle_id	INT	FK → Vehicle(id), NOT NULL

Column	Data Type	Constraints
customer_id	INT	FK → Customer(customer_id), NOT NULL
price	DECIMAL(12,2)	NOT NULL, CHECK (price > 0)
date	DATE	NOT NULL
status	VARCHAR(20)	CHECK (status IN ('Pending', 'Completed', 'Cancelled'))

Indexes

- INDEX idx_purchase_customer (customer_id)
- INDEX idx_purchase_vehicle (vehicle_id)

9) Payment

Column	Data Type	Constraints
payment_id	INT	PK, AUTO_INCREMENT
booking_id	INT	FK, Rental_Booking(booking_id)
purchase_id	INT	FK, Purchase(purchase_id)
payment_type	VARCHAR(20)	CHECK (payment_type IN ('Rental', 'Purchase'))
amount	DECIMAL(12,2)	NOT NULL, CHECK (amount > 0)
method	VARCHAR(20)	CHECK (method IN ('Cash','Card','Online'))
isPaid	BOOLEAN	DEFAULT FALSE

Additional Constraint

- CHECK: (booking_id IS NOT NULL OR purchase_id IS NOT NULL)

Indexes

- INDEX idx_payment_booking (booking_id)
- INDEX idx_payment_purchase (purchase_id)
- INDEX idx_payment_paid (isPaid)

10) User

Column	Data Type	Constraints
user_id	INT	PK, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
role	VARCHAR(30)	NOT NULL
email	VARCHAR(100)	UNIQUE
hire_date	DATE	NOT NULL
isActive	BOOLEAN	DEFAULT TRUE

Indexes

- UNIQUE(email)
- INDEX idx_user_role (role)

Permissions:

I. System Administrator

- Full CRUD on all tables
- Manage users & roles
- Override bookings
- View financial reports

II. Rental Manager

- Manage Vehicles
- Approve/Cancel Bookings
- Assign Drivers
- View Maintenance & Insurance
- View Payments (read-only)

III. Customer Service Representative

- Create bookings
- Update booking status
- View customer details
- Generate invoices
- Cannot delete vehicles or payments

IV. Accountant (Optional Additional Role)

- Manage Payments
- View Purchase records
- Financial reporting
- Cannot modify bookings or vehicles

Summary of Constraints Used

- Primary Keys on all tables
- Foreign Keys with referential integrity
- UNIQUE on emails, license numbers
- CHECK for:
    - Status fields
    - Date validity
    - Positive monetary values
    - Enum-like fields
- Boolean defaults for flags